Chapter 3

Cardiac anesthesia

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Monitoring, and respiratory and cardiovascular support are discussed in Chapter 5—Cardiac ICU (📖 pp157–252)

Basic principles of anesthesia

Anesthesia may be general, regional, or local. The vast majority of cardiac surgery is performed under general anesthesia.

General anesthesia

A general anesthetic has three components: *hypnosis* (reversible state of unconsciousness), *analgesia*, and *muscle relaxation*

Hypnosis

- Hypnosis has three phases: *induction*, *maintenance*, and *emergence*.
- Induction and maintenance can be by iv or inhaled drugs.
- Intravenous induction is faster than inhalational induction.
- Some of the iv drugs used for hypnosis are the same as those used for analgesia—just in higher doses.
- As the hypnotic effect of iv anesthetic drugs is short, iv maintenance is by continuous infusion.
- Most induction agents are associated with a degree of *cardiorespiratory depression*, and usually cause *vasodilation*: particular care must be taken when inducing patients with acute tamponade, or critical AS to avoid acute decompensation requiring emergency intervention.

Analgesia

- Analgesia reduces both somatic and autonomic response to pain.
- Analgesics used in general anesthesia are usually opiates.
- Opiate analgesics are associated with respiratory depression.

Muscle relaxation

- Muscle relaxants are classified as depolarizing or non-depolarizing.
- Non-depolarizing muscle relaxants competitively bind with acetylcholine receptors at the neuromuscular junction and last up to 60min.
- They are reversed by *neostigmine*, an anticholinesterase.
- Depolarizing muscle relaxants non-competitively depolarize the muscle end plate causing *fasciculation* and rapid onset paralysis.
- They are effective for 5–15min.
- Choice is dictated by duration and cardiovascular effects (vecuronium and rocuronium have no cardiovascular side effects).

Examples of anesthesia for cardiac surgery

Regime A

- Premed: temazepam 10–20mg po 1h preoperatively.
- Induction: fentanyl 5–10 micrograms/kg and etomidate 10–20mg.
- Maintenance: isoflurane 1–2%, or further boluses of fentanyl.
- Muscle relaxant: pancuronium 0.1mg/kg.

Regime B Total intravenous anesthesia (TIVA)

- Premed: hyoscine 0.3mg im and 5mg morphine im 1h preoperatively.
- Induction: propofol 4–10 micrograms/kg/h.
- Maintenance: propofol 4–10 micrograms/kg/h + remifentanyl 0.3 micrograms/kg/min.
- Muscle relaxant: rocuronium 0.6–1mg/kg.

Fasting

- Elective patients should be fasted for 6–8h preoperatively to allow gastric emptying (NG feeds should be stopped 6h pre-op), reducing the chances of regurgitation and aspiration, although it is ok to take oral medication.
- Pain and opiates slow gastric emptying.
- Rapid sequence induction is used in an emergency to minimize the risk of gastric aspiration.
- Be explicit about fasting to nursing staff otherwise your outliers may get breakfast on the morning of surgery.

Preoperative assessment

- In addition to assessment that covers much of the ground described in Chapter 1 the anesthesiologist assesses the patient with particular regard to adverse reaction to general anesthesia, family history of malignant hyperpyrexia, dentition, and airway.
- The anesthesiologist explains the anesthetic plan, monitoring, induction and intubation, the possibility of awareness during surgery, and emergence while intubated on ITU, and potential complications.

Airway assessment

Ease of intubation has been graded according to the best possible view obtained on laryngoscopy. Grades 3 and 4 are difficult intubations. Difficult intubations can usually be predicted in advance by the presence of a combination of signs. In the ***Mallampati scoring system*** (📖 p644) the patient sits with mouth open and tongue protruded opposite the anesthesiologist, who assesses the structures visible at the back of the mouth. In Class 1 patients the faucial pillars, soft palate, and uvula are visible. In Class 4 patients none of these soft tissues are visible: these patients are extremely likely to be difficult intubations. In addition to the Mallampati scoring system the following predict difficult intubation:

- Thyromental distance <3 finger breadths with the neck extended.
- More than 2 Wilson risk factors; obesity, restricted head and neck movement, restricted jaw movement, receding mandible, buck teeth.
- Inability to flex the chin onto the chest.
- Reduced jaw opening (<2 finger breadths).
- Radiological features including mandibular depth.

Scoring preoperative condition

Although risk scoring for cardiac surgery has its own dedicated systems (Euroscore and STS Score, 📖 Inside cover) all surgical patients are also allocated an ASA grade (Box 3.1) preoperatively.

Box 3.1 American Society of Anesthesiologists (ASA) score

Grade I: healthy patient.
Grade II: mild systemic disease, no functional limitation.
Grade III: moderate systemic disease, significant functional limitation.
Grade IV: severe systemic disease that is a constant threat to life.
Grade V: moribund patient unlikely to survive 24h.

Conduct of anesthesia

Premedication

There is wide variation in choice of premedication, which is normally given an hour before the patient is transferred to the surgery area. Options include:

- A benzodiazepine, e.g. temazepam 10mg po, or lorazepam 2–4mg po: lorazepam has a half-life of 10–20h, temazepam is shorter acting.
- Intramuscular morphine 0.1mg/kg and scopolamine 4–8 micrograms/kg.
- Intramuscular morphine 0.1mg/kg and hyoscine 0.3 mg.

Transfer

Prior to transfer to the surgical area the following are verified:

- The patient's identification in the form of a wristband with name, date of birth, and a hospital number.
- The patient's name, date of birth, hospital number, operation details, signature, and date of signing, on the consent form.
- Availability of and patient details on cross-matched blood.
- Presence of responsible surgeon.

Prior to induction

The anesthesia area is prepared well in advance of the patient's arrival to ensure that all necessary equipment is present and working, and that the specific drugs required for induction are drawn up and labelled. Practice varies, but most anesthesiologists establish all monitoring, with the exception of the central venous catheter, prior to induction, as this ensures that any hemodynamic or respiratory compromise occurring as a result of induction will be rapidly detected and treated.

- The patient is sat at 45°.
- Oxygen is given by face mask.
- A pulse oximeter is attached and oxygen adjusted appropriately.
- EKG leads attached, checked and II and V5 used for display.
- A 14-gauge venous cannula is placed at the left wrist under local anesthesia, unless the left radial artery is to be harvested.
- A radial arterial catheter is placed under local anesthesia, transduced and a baseline blood gas is measured.

Patients at high risk of hemodynamic instability

Patients with the following lesions are at high risk of hemodynamic decompensation during the induction sequence, and a surgeon should be immediately available, and in the case of acute tamponade the patient prepped and draped ready for incision prior to induction:

- Tight LMS stenosis.
- Severe aortic and mitral stenosis.
- Acute cardiac tamponade or mediastinal bleeding.
- Severe pulmonary hypertension.
- Very poor LV function or cardiogenic shock.
- Pregnancy.

Induction

There is a wide choice of induction agents, and techniques. Use of fentanyl is described here: fentanyl has all the advantages of morphine (overdose is rare, it is not a myocardial depressant, it is non-explosive compared to inhalational agents, it facilitates postoperative management) as well as being a short-acting agent that is safe in the management of critically ill patients. The disadvantage is a risk of awareness.

- The patient is placed flat on a single pillow.
- Preoxygenation with 100% O_2 by face mask helps to avoid pulmonary hypertension in at-risk patients, as well as giving a safety margin by filling the functional residual capacity with O_2 if there are any problems establishing a secure airway later.
- 1mg of pancuronium may help avoid narcotic-induced rigidity.
- Induction: fentanyl 5–10 micrograms/kg and etomidate 10–20mg iv.
- As soon as the patient is unresponsive an appropriate dose of muscle relaxant up to a maximum of 0.1mg/kg of pancuronium is given.
- It may be necessary to give volume or a vasopressor such as metaraminol to treat hypotension at this stage.
- The trachea is intubated, and the ET tube secured.
- The table is placed in Trendelenburg position so that the central venous catheter, and pulmonary artery catheter if indicated, can be inserted.
- The urinary catheter is inserted, and the patient shaved.
- The table is levelled, and the pressure transducers adjusted.
- Prophylactic antibiotics are given: vancomycin is run an hour before incision to maximise serum concentrations.
- The EKG leads for the defibrillator and any TEE probe are placed, and an electrocautery pad is placed.
- The patient's arms are carefully secured to avoid compression injury.

Problems during induction

Respiratory

- Failure to secure an airway.
- Difficult intubation.
- Aspiration.
- Bronchospasm.
- Pulmonary hypertensive crisis.

Cardiovascular

- Cardiac arrest (critical AS and/or LMS, acute tamponade).
- Hypotension or hypertension.
- Ischemia.
- Arrhythmias.

Other

- Anaphylaxis
- Complications of with invasive monitoring (📖 p173, 175, 177).
- Dental and oropharyngeal trauma.
- Brachial plexus, ulnar and radial nerve injury, occipital alopecia, heel and sacral tissue necrosis may result from improper positioning.

Pre-bypass anesthesia

Key facts

Anesthesia management is aimed at optimizing several parameters over a period of time when surgical activity results in variable levels of sympathetic stimulation, as well as manipulation that impact cardiac function, namely:

- The ratio of myocardial oxygen supply: demand to prevent ischemia.
- Preload and afterload, particularly in patients with valvular defects.
- Heart rate, rhythm, and ventricular contractility
- Systemic and pulmonary vascular resistance, particularly in patients with pulmonary hypertension or congenital heart disease.

Median sternotomy

Skin incision, sternal division, sternal retraction, and dissection of the adventitia over the aorta result in high levels of sympathetic stimulation. An adequate depth of anesthesia is necessary to avoid tachyarrhythmias and hypertension. The surgeon should check that the anesthesiologist is happy with the level of anesthesia before starting. Sternotomy occasionally results in significant blood loss. Some surgeons ask for the lungs to be deflated prior to sternal division to reduce the minimal risk of damage to the lung parenchyma. A baseline ACT may be measured at this point.

Redo-sternotomy

This is discussed on 📖 pp344–347. The main differences in redos are:

- The patient must have ***external defibrillator pads*** on.
- Check cannulation and conduit plan with the surgeon as this dictates possible sites for siting arterial lines (e.g., patient will need left radial line if right axillary artery cannulation or radial harvest planned).
- There is a risk of injuring grafts, ventricle, or great vessels on sternal division: in high-risk cases the patient will be cannulated and even placed on bypass prior to resternotomy to reduce the risk.
- The dissection that must take place before the heart can be cannulated is longer, and involves more mechanical disturbance and bleeding.

Conduit harvest

- This is a period where sympathetic stimulation is minimal, and the anesthesiologist may need to treat bradycardia and hypotension.
- Compression of the left subclavian artery by sternal retractors may mean left radial lines do not accurately reflect systemic pressures.
- Major ***occult blood loss*** may occur into the pleurae or the leg wound.
- Brachial plexus injury can result from overabduction of the arm when positioning it for radial artery harvest.
- For harvesting the IMA the table needs to be raised, sometimes tilted away from the surgeon, and sometimes the lungs deflated temporarily.
- Heparin (300 units/kg iv) is given before the IMA is divided otherwise it could thrombose: the surgeon will ask for heparin to be given and the anesthesiologist should acknowledge the request.
- Check the ACT 3min after heparin is given (📖 p93).
- If the surgeon injects papaverine directly into the IMA it may be necessary to give phenylephrine to counteract vasodilation and hypotension.

Heparin and protamine: avoiding catastrophic drug errors

Principles of bypass are discussed in detail on 📖 pp102–124.

- Acknowledge all communication, and repeat back key instructions e.g., 'give protamine' verbatim.
- The ACT must be >400 before bypass.
- Protamine must *never* be given while the patient is on bypass.

Preparing to cannulate

- Hitching the pericardium to the sternal edges frequently leads to transient hypotension as a result of vagal stimulation, and mechanical constriction of the SVC and IVC: if this does not respond to adequate volume resuscitation the surgeon should 'drop' the pericardium. TEE is helpful to reassess valve and ventricular function, guide coronary sinus catheter (Fig. 3.1) and assess aorta.
- Handling the RA while placing purse-strings may result in AF: ***synchronized DC cardioversion*** 10–50J with internal paddles is safest when patient is heparinized and cannulated as VF is occasionally induced.
- Blood pressure should ideally be <100mmHg systolic at aortic cannulation to reduce the risk of aortic dissection and bleeding.
- Once the aorta is cannulated and pipe connected, volume can be given rapidly by the perfusionist if necessary. Blood loss can be drained by pump suckers directly back to the pump if the ACT is >400.
- Once the venous cannula is in and connected the patient can be put on bypass if needed, but in order to minimize bypass time the surgeon will usually check condut and site additional vents first.

Management of conditions which affect bypass

Antithrombin III deficiency

If ACT remains <400 despite large heparin dose, give FFP to treat antithrombin III deficiency and recheck ACT.

Pregnancy

Although maternal mortality is the same as for a non-pregnant patient, fetal mortality approaches 50%. Placental ischemia occurs as a result of microemboli, elevated IVC pressure which reduces venous drainage, and pump flows inadequate for the hyperdynamic circulation associated with pregnancy. Uterine blood flow is not autoregulated. Dilution of progesterone may result in induction of labour. Management is targeted at establishing adequate pump flows, and avoiding hypoxia and hypothermia.

Heparin-induced thrombocytopenia

Heparin substitutes such as danaparoid may be used (📖 p288).

Sickle cell

Cardiopulmonary bypass exacerbates the tendency of erythrocytes to sickle because of vasoconstriction due to hypothermia, stasis, hypoxia, and acidosis. Even patients with sickle cell trait are at risk of a bypass-induced sickle cell crisis. This condition is managed with preoperative partial exchange transfusion to reduce Hb-S from 100% to less than 33%. Hypoxemia, acidosis and dehydration must be avoided preoperatively.

Anesthesia on bypass

Maintenance of bypass

Blood concentrations of drugs are diluted by the prime in the bypass circuit, and additional muscle relaxant and anesthetic agent may be required on bypass. Once full flow is achieved the patient is disconnected from the ventilator (just switching off the ventilator can result in air trapping, leaving the lungs inflated, interfering with the operative field).

Routine checks during bypass

- ***Anticoagulation:*** the ACT is checked after initiating bypass and every 30min thereafter, with additional heparin given in 5000-unit increments if necessary to maintain the ACT >500.
- ***Blood gas and acid base status:*** this is checked after initiating bypass and every 30min thereafter, with the PaO_2 maintained at >14kPa (105mmHg) by adjusting FiO_2 to the oxygenator, and pH adjusted according to whether management is pH stat or alpha stat (📖 p124).
- ***EKG:*** checked for inappropriate electrical activity during arrest and ischemic changes and arrhythmias pre-and post-arrest.
- ***CVP and MAP:*** these are recorded at 5–10min intervals, and should be 1–10mmHg and 50–70mmHg respectively (arterial pressures are maintained 70–80mmHg in elderly/arteriopathic patients).
- ***Urine output:*** the bag is usually emptied before bypass so that total urine output during bypass can be measured.
- ***Face:*** periodically check that there is no facial edema, or asymmetry in colour or temperature, and no pupil asymmetry, due to misplacement of the venous or aortic cannulas (📖 p110). Sweating, lacrimation, mydriasis may indicate inadequate depth of anesthesia.
- ***Core and peripheral temperature:*** nasopharyngeal or tympanic temperature probes measure core temperature which the perfusionist maintains at the temperature requested by the surgeon.

Common abnormalities on bypass

Problems of high line pressures, poor venous drainage and poor electromechanical arrest are discussed on 📖 pp113, 123. Management of catastrophic bypass problems is described on 📖 pp116–119.

Hypotension

- Flow rate deliberately decreased by perfusionist, e.g., when surgeon clamping aorta or dealing with bleeding.
- Inadequate flow as a result of inadequate arterial cannula size for patient's BSA, kinked or clamped arterial line, poor roller head occlusion.
- Low SVR due to vasodilation (give phenylephrine) or hemodilution.
- Cytokines in sudden return of pooled blood via pump suckers.
- Transducer error, calibration error, tubing disconnection, kinking or compression of subclavian, brachial or radial artery.
- Arterial cannulation problems including selective cannulation of carotid arteries, aortic dissection, or rarely reversed cannulation.

Hypertension
- High SVR due to vasoconstriction mediated by endogenous catecholamines, hypothermia, awareness, and hyperoxia: treat the likely cause.
- Pump flows deliberately increased by perfusionist, e.g., to speed rewarming.
- Selective cannulation of subclavian proximal to radial artery catheter.
- Transducer too low.

High CVP
Poor venous drainage (📖 p113), inadvertently snaring SVC (📖 p111) proximal to cannula, obstructing venous return.
- Catheter abutting SVC wall, venous pipe or snared by surgeon.

Slow nasopharyngeal cooling or rewarming
- Excessive vasoconstriction.
- Low pump flows.
- Temperature probe misplaced or malfunctioning.
- High CVP impeding cerebral perfusion pressure gradient.
- Carotid artery malperfusion (e.g., inadvertent selective cannulation).
- Rarely increased intracranial pressure.

Hypoxemia
- Inadequate FiO_2 to the pump oxygenator.
- Low pump flows.
- Pump oxygenator malfunction (📖 p117).
- Causes of low SvO_2 are as given earlier, and $\uparrow O_2$ consumption secondary to hyperthermia, shivering, and rarely malignant hyperpyrexia.

Oliguria
- Post-renal problems such as kinked catheter tubing, misplacement, clamping, and catheter occlusion by gel or clot.
- Decreased glomerular filtration rate as a result of decreased renal perfusion due to inadequate pump flows, hypotension, renal vasoconstriction by drugs and hypothermia, increased IVC pressures, non-pulsatile perfusion, clamping of descending aorta and circulatory arrest.
- Pre-existing renal impairment.

Hemoglobinuria
- Hemolysis due to pump and pump sucker trauma.
- Trauma to urinary tract in setting of heparinization.
- Blood transfusion reaction.
- Rarely contamination of pump circuit with water from heat exchanger.

Electrolyte imbalance
- $\uparrow K^+$ due to administration of cardioplegia, impaired renal function.
- $\downarrow K^+$ due to dextrose and insulin administration, polyuria.

Awareness
About 1% of patients have recall of intraoperative events. This is due to a combination of dilution of anesthetic agents by the pump prime and absorption of fentanyl into the prime circuit: some patients are too unwell to tolerate higher doses of anesthetic agent. It is commonest when the patient is warm, as hypothermia induces unconsciousness. Variation between individuals means there is no dose of drug that guarantees unconsciousness. Careful titration of anesthetic agents is critical.

Anesthesia post-bypass

Criteria for discontinuing bypass

- No conditions that require CPB, e.g., residual valve leak, bleeding.
- Satisfactory rhythm, ventricular rate >60, paced if necessary, with adequate cardiac function, and minimal intracardiac air.
- Nasopharyngeal temperature 36.5–37.5°C.
- K^+ 4.0–5.0.
- pH 7.30–7.50.
- pO_2 >16kPa (120 mmHg).
- Anesthesiologist has resumed ventilating the lungs.

Weaning from bypass

Weaning from bypass is discussed in more detail on 📖 pp114–115. The anesthesiologist and perfusionist ensure that all parameters are suitable for discontinuing bypass without delay once surgery is completed.

The usual sequence of events for the anesthesiologist is:

- The perfusionist begins rewarming during the last surgical anastomosis. Core temperature should be 36.5–37.5°C as the patient loses heat once off bypass; overshooting may be bad for neurocognitive function.
- Place the table in Trendelenburg position for de-airing.
- Hand ventilate carefully ensuring the IMA graft is not placed under tension by lung inflation, and checking that the lungs expand fully.
- Place the patient on the ventilator. Surgeon de-airs heart (📖 p114).
- The perfusionist gradually allows the heart to fill and eject by progressively occluding the venous line at the same time as reducing pump flows, until pump flow ceases and the patient is 'off bypass'.
- Bypass can be terminated immediately at any point during this procedure, and it is usually better to come off relatively underfilled as excessive preload and afterload increase myocardial oxygen demand at the time when the heart is least able to deal with this.
- Rapid, controlled transfusion is given by perfusionist in aliquots of 50–100mL from the pump via the arterial line on request, and pump suckers can still be used to return blood to the pump, so there is no need to give volume intravenously.
- Inotropes, chronotropes, vasodilators, or vasoconstrictors are titrated.
- If hemodynamic instability or evidence of ischemia it may be necessary to return go back on bypass for a period to reduce the demand on the heart, and allow it to recover from stunning.
- Failure to wean from bypass is discussed on 📖 p117.
- Protamine slow push (3mg/kg/iv or dose calculated by Hepcon®) is given to reverse heparin ***after*** bypass is finally terminated.
- The surgeon will spend a period of time checking hemostasis while protamine is given, allowing time for full reversal of heparin.
- The surgeon decannulates the aorta once the pump is empty or the heart is full: systemic pressure should be <110mmHg systolic to reduce the risk of hemorrhage or dissection at this point.

Giving protamine

Management of ***protamine reactions*** is described on 📖 p116. Once the surgeon is sure that bypass will not be required again they will ask the anesthesiologist to give the protamine. It is good practice not to draw up protamine until this request is made as bypass has been inadvertently and catastrophically terminated by giving protamine by mistake instead of a drug in a nearby syringe. Although bypass is terminated blood can still be transfused to the patient from the pump via the arterial line if required, but use of pump suckers should be discontinued once protamine starts.

Avoiding catastrophic pump failure

- Do not draw up protamine until asked to give it.
- Never give protamine while the heart is on bypass.
- State clearly when you start to give the protamine: pump suction *should* be discontinued immediately by perfusionist or surgeon.

Variations in sequence

Decannulating before protamine is given

- If the surgeon expects that attempting to decannulate the aorta may cause sufficient hemodynamic compromise that bypass must be reinstituted, they may elect to decannulate before protamine is given. Some surgeons routinely decannulate the aorta before protamine is given to eliminate any risk of thrombus formation at the tip of the cannula.

Placing the aortic cannula into the right atrium after decannulation

If the surgeon anticipates substantial bleeding from the aortic cannulation site, they may elect to place the arterial cannula into the RA via the purse-string previously used for the venous cannula, so that blood transfusion from the pump may be continued if required.

Surgical maneuvers that cause hypotension

Always look at the operative field before treating hypotension: the cause is usually surgical and temporary.

- Aortic decannulation: (1) surgeon may compress heart or occlude venous inflow to deliberately lower arterial pressure in order to decannulate safely, (2) if significant bleeding occurs.
- Oversewing the atrial purse-string if arrhythmias or bleeding result.
- Lifting the heart to inspect anastomoses.
- Packing the pericardium.
- Hitching the pericardium: most surgeons unhitch at this stage.
- Approximating the sternal edges if there are packs around the heart.
- Deliberate occlusion of the IVC by the surgeon to reduce BP rapidly (see aortic decannulation).
- Pacing problems: use of electrocautery while on demand pacing if slow (or no) underlying rhythm, deliberate or inadvertent disconnection.
- Use of electrocautery interfering with IABP in EKG mode.

Anesthesia for off-pump surgery

Key facts

While off-pump surgery confers the benefits listed on 📖 p305, it presents surgeon and anesthesiologist with two major challenges, which anesthetic and surgical technique must overcome:

- Reduction in cardiac output when positioning the heart.
- Interruption to coronary blood flow during each distal anastomosis.

Anesthetic technique

One important reason for choosing off-pump surgery is that patients can sometimes be extubated earlier, but this can only be accomplished with appropriate choice of anesthetic agents and analgesia.

- Smaller doses of short-acting benzodiazepines are given as premedication, which may be supplemented by midazolam if necessary.
- Fentanyl and remifentanil are shorter-acting anesthetic agents that may be used in preference to propofol and opioids.
- Lines inserted by the anesthesiologist are the only means of giving rapid transfusion in the absence of an aortic cannula: there *must* be adequate wide-bore venous access.
- To help maintain cardiac output when the heart is positioned, patients are actively hydrated.
- Heparin is frequently given at a lower dose but check with the surgeon: doses vary between 100–300 units/kg iv.
- Most units routinely use TEE to assess LV function intraoperatively, and routinely place PA catheters to help detect ischemic changes (e.g., ischemic MR, elevated PA pressures).
- Continual measurement of ST segment changes is a very useful adjunct.
- Make sure that a sterile warming blanket is placed on the patient, as there will be no other way of securing normothermia intraoperatively.

Hemodynamic management during anastomoses

Hemodynamic changes occur rapidly, but can be anticipated by knowing the sequence of surgery, as well as by adjuncts to standard monitoring: ST segment analysis, TEE, and cardiac output measurements. Pressor support may be required during the distal anastomosis, during which preload must be optimized. The systemic pressure needs to be relatively high prior to the distal anastomoses, as positioning the heart inevitably results in a decrease in cardiac output. For the proximal anastomoses the systemic pressures need to be lower, to minimize the risk of aortic tears and dissection. It is possible to perform the proximal anastomoses first: this may be indicated to minimize the period of relative ischemia, and exploit low systemic pressures. The use of intracoronary shunts helps to reduce myocardial ischemia while performing distal anastomoses. The use of TEE may allow the anesthesiologist to assess whether the heart is likely to tolerate particular positions, by identifying valvular regurgitation, and ventricular impairment before the heart decompensates. Tachycardia makes distal anastomoses more difficult. It is easier to maintain hemodynamic stability when the pH is physiological. BP and cardiac output should return to baseline between each anastomosis.

Respiratory management

Sometimes it is necessary to stop ventilating for short periods to optimize surgical exposure. This is usually during anastomoses to the obtuse marginal vessels. The table will also be placed in steep Trendelenburg position for the right-sided anastomoses, which results in a reduction in pulmonary compliance and functional residual capacity. This may result in hypoxia, hypercarbia and atelectasis, which should be addressed by careful hyperventilation.

Postoperative management and extubation

Not all surgeons choose to reverse heparinization with protamine. If the patient is normothermic, not acidotic, and adequately ventilating, hemodynamically stable and not bleeding once the sternum is closed, they are candidates for waking and extubation in theatre, or shortly within intensive care. Adequate analgesia is imperative. Although intrathecal morphine has been used in some centres, most units use patient- and nurse-controlled analgesic iv morphine infusions. High O_2 flows, bronchodilator therapy, and chest physiotherapy are important adjuncts. Patients with good ventricular function and a full revascularization are usually hypertensive and tachycardic. They do not produce a diuresis as they are often relatively underfilled. A significant metabolic acidosis frequently develops during the first 6h postoperatively, resolving within 12h. The cause of the acidosis is unclear but may reflect an intraoperative low cardiac output state. As bypass cannot be used to warm these patients they may be hypothermic.

- Ensure these patients are adequately filled.
- Use warming blankets and infusion warmers to achieve normothermia.

These patients sometimes receive a reduced dose of intraoperative heparin depending on surgical preference, and should therefore receive early aspirin (75mg po) and low-molecular-weight heparin (5000 units sc) to ensure graft patency and avoid thromboembolic complications.

Epidural anesthesia in cardiac surgery

This technique is also used in cardiac surgery requiring cardiopulmonary bypass. An epidural is sited at either T1/2, T2/3, or T3/4. This may be done as early as the night before surgery. 8mL 0.5% ropivacaine with 20 micrograms of fentanyl is infused before induction of anesthesia so that sensory spread can be assessed. After induction the infusion is continues at 5–15mL/h. A sensory blockade of T1–T10 can be achieved. The epidural is discontinued on postoperative day 3. The most important complication is ***paraplegia*** from a spinal cord haematoma (risk approximately 1:3000). The benefits of this technique include:

- Large improvement in pain relief.
- Reduced opioid and NSAID requirement.
- Improved physiotherapy compliance.
- Improved respiratory function.
- Reduced length of ITU and hospital stay have been claimed.
- Reduced incidence of depression.

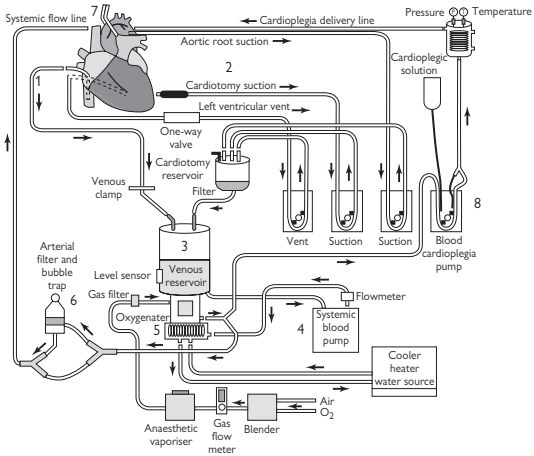


Fig. 3.1 Standard cardiopulmonary bypass circuit. Venous cannulae (1), vents and pump suckers (2) drain blood from the patient into the venous reservoir (3) which is then pumped (4) through an oxygenator (5) and then filtered (6) and returned at arterial pressure into the arterial cannula (7) (aorta, femoral or axillary).

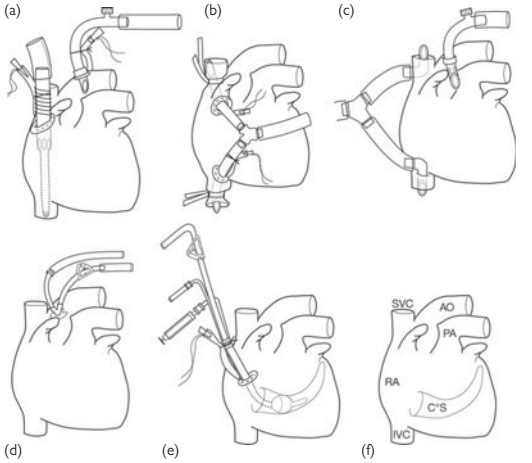


Fig. 3.2 Types of cannulation. (a) Aortocaval cannulation with 2-stage atrial cannula (note 2 levels of perforations, one at IVC and one within RA). (b) Bicaval cannulation. (c) Direct bicaval cannulation with right angled venous cannula. (d) Antegrade cardioplegia cannulae with root vent. (e) Retrograde cardioplegia cannula with syringe of saline to inflate balloon and manometry line. (f) Labelled heart: superio vena cava (SVC), pulmonary artery (PA), aorta (AO), right atrium (RA), inferior vena cava (IVC).